

Manipulation of Formula Worked Examples - Advanced

Example 1

Consider $m \neq 0$. Make v the subject of the expression:

$$K = \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = K$$

$$mv^2 = 2K \quad \text{Multiply both sides by 2}$$

$$v^2 = \frac{2K}{m} \quad \text{Divide both sides by } m$$

$$v = \sqrt{\frac{2K}{m}} \quad \text{Square root both sides}$$

Example 2

Make b the subject of the expression:

$$\frac{p}{q} = \sqrt{\frac{(a+2b)}{(a-2b)}}$$

$$\sqrt{\frac{(a+2b)}{(a-2b)}} = \frac{p}{q}$$

$$\frac{(a+2b)}{(a-2b)} = \frac{p^2}{q^2} \quad \text{Square both sides}$$

Cross multiply to obtain:

$$(a + 2b)q^2 = (a - 2b)p^2$$

$$aq^2 + 2bq^2 = ap^2 - 2bp^2 \quad \text{Removing the brackets}$$

$$2bp^2 + 2bq^2 = ap^2 - aq^2 \quad \text{Bring the } b \text{ elements to one side of the expression}$$

$$2b(p^2 + q^2) = a(p^2 - q^2) \quad \text{Isolate the } 2b \text{ and the } a$$

$$b = \frac{a(p^2 - q^2)}{2(p^2 + q^2)} \quad \text{Divide both sides by } 2(p^2 + q^2)$$